Charles Zhao

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WORK EXPERIENCE

Lawrence Livermore National Laboratory **Summer 2023 & 2024**
Cybersecurity Intern, 40 hrs/wk *Livermore, CA*

- Wrote Python unit and integration tests using pytest, sqlite3, and psycopg2, reducing debugging effort and validating application compatibility with the CI/CD pipeline.
- Built a dynamic web application for the Department of Homeland Security using the MEAN stack (MongoDB, Express.js, Angular, Node.js), delivering a scalable and efficient solution for enhanced data management.
- Developed a Python script to automate report parsing and identify suspicious IP addresses and domains interacting with the network, improving threat detection and security monitoring.

PROJECTS

KubeRange **Mar – Apr 2026**
- Developed a full-stack web application (React, FastAPI, PostgreSQL) hosted on Kubernetes that dynamically provisions and tears down isolated vulnerable web app instances (DVWA, Juice Shop) on demand, with an integrated ELK stack for data collection and observability.

Cascading Detection Engine **Feb – Apr 2026**
- Built a multi-stage phishing detection engine combining rule-based analysis, SVM, Isolation Forest, and DistilBERT, escalating ambiguous samples between stages and using logistic regression for final score fusion.

Cybersecurity Daily Digest **Feb – Apr 2026**
- Built an automated cybersecurity news pipeline using RSS feeds, GitHub Actions, and an LLM to collect, summarize, and deliver a daily digest.

Domain Sinkhole & Network Traffic Analysis **Nov – Dec 2025**
- Deployed Pi-hole on a Raspberry Pi as a local DNS sinkhole to filter and block unwanted domains across all devices on a home network. Investigated captured DNS queries using Wireshark and documented findings on GitHub.

Custom C++ Stager **Aug 2024 – May 2025**
- Developed a custom C++ stager within a sandboxed Windows environment to study and evaluate modern antivirus evasion techniques, including process injection and dynamic API resolution.

Malicious File/URL Scanner **Jan – May 2024**
- Developed a Python microservice hosted on Proxmox that scanned email attachments and URLs using the VirusTotal API and database to identify malicious content.
- Validated detection capabilities in an isolated virtual environment using known malicious files and URLs from public sources.

EDUCATION

Virginia Tech
- **Master of Engineering, Computer Engineering – May 2025 | GPA: 4.0**
- **Bachelor of Science, Computer Engineering, Magna Cum Laude – May 2024 | GPA: 3.61**
  Programs: SFS CyberCorps Scholar; IC CAE

CERTIFICATIONS & SKILLS

- **Certifications:** CompTIA Security+; Cranium AI Red Team Certification
- **Skills:** Python; C++; Linux; Kubernetes; Proxmox; Wireshark; Burp Suite; Testing & Automation; Ghidra